

Assignment 4

Name

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Course Details : SECR2033-02

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Question 1

Word size = 32 bits

Block size 4K words = $4 \times 1024 = 4096$ words

i) $1024 \times 4K$ words
 $= 4M$ words

$4M$ words $\times 32$ bits = 128 Mbit

$128 \text{ Mbit} \div 8 = 16 \text{ MB}$

ii) $\frac{18 \text{ Mbit}}{32 \text{ bits}} = \frac{8 \times 1048576 \text{ bits}}{32} = 262144 \text{ word}$
 $= 256 \text{ K words}$

$= \frac{256 \text{ K words}}{4 \text{ K words}}$

$= 64 \text{ blocks}$

$$\begin{aligned} \text{(i) Memory Capacity} &= 64 \times 1\text{M} \\ &= 2^6 \times 2^{20} \\ &= 2^{26} \end{aligned}$$

$$\begin{aligned} \text{(ii) } 2^n &= 2^{26} \\ n &= 26 \text{ bits} \end{aligned}$$

$$\text{(ii) Total sets} = \frac{32\text{K}}{2} = 16\text{K}$$

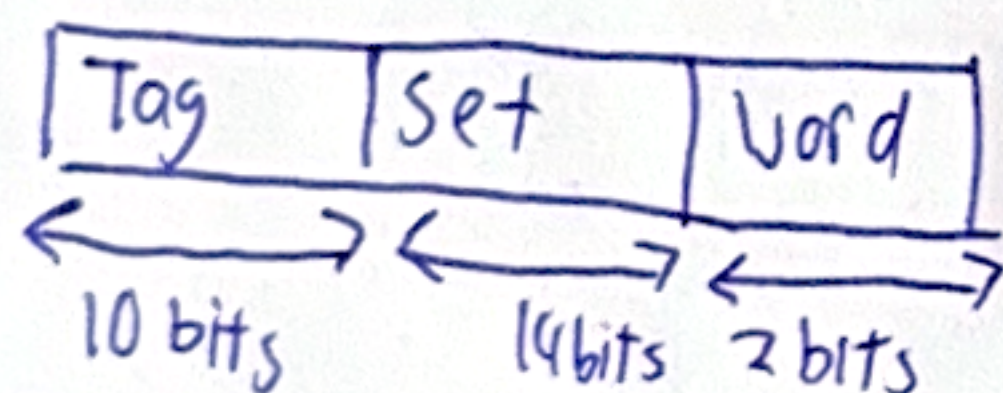
$$\begin{aligned} 16\text{K} &= 16 \times 1024 \\ &= 16384 \text{ sets} \end{aligned}$$

$$\begin{aligned} 16384 &= 2^4 \times 2^{10} \\ &= 2^{14} \end{aligned}$$

$$\begin{aligned} \text{(iii) Word bits} &= \log_2(4) \\ &= 2 \text{ bits} \end{aligned}$$

$$\text{Set bits} = 14 \text{ bits}$$

$$\begin{aligned} \text{Tag bits} &= \text{Total Address Bits} - (\text{Set bits} + \text{Word bits}) \\ &= 26 - (14 + 2) \\ &= 10 \text{ bits} \end{aligned}$$

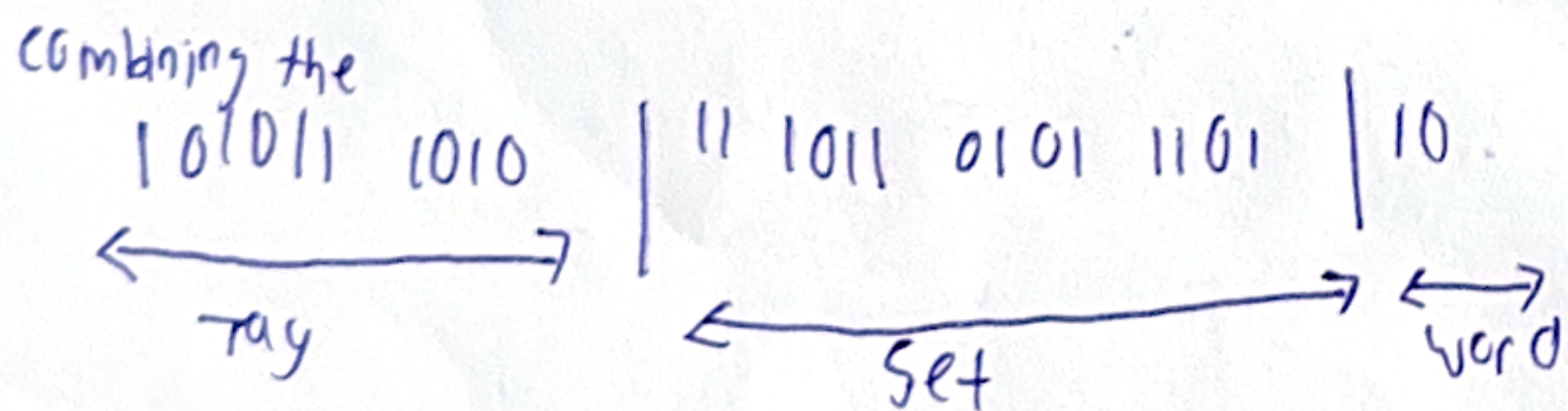


$$\text{(iv) Tag: } 2\text{BAh} \rightarrow 10 \ 1011 \ 1010_2 \text{ (10 bits)}$$

$$\text{Set: } 3\text{B5Dh} \rightarrow 11 \ 1011 \ 0101 \ 1101_2 \text{ (14 bits)}$$

$$\text{word: } 10 \text{ (2 bits)}$$

Combining the



0010	1011	1010	1110	1101	0111	0110
2	B	A	E	D	7	6

main memory address = 2BAED76